Power, Compute, and Geography: A Quantitative Framework for AI and Cloud Infrastructure Competitiveness

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https://github.com/finnrcase/ai-infrastructure-economics

https://ai-infrastructure-economics-mnhgqsnwhjshnszfkdoozq.streamlit.app/

Abstract

The need to power artificial intelligence is increasing demand for computing infrastructure, creating new challenges within energy markets, electricity systems, and region based economic deployment. Current research and literature focuses on environmental impacts of AI or projected electricity demand growth, while far less attention is given to economic and geographical factors that determine where and how AI energy infrastructure is built.

This paper develops a quantitative framework to evaluate AI infrastructure competition across regions within the United States. The framework incorporates six state-level variables: electricity pricing, grid capacity, renewable energy penetration, existing data center infrastructure, behind-the-meter systems potential, and interconnection friction. These variables are used within the regional competitiveness model to compare 15 states that were found to be viewed as leading markets or potential markets for future AI infrastructure deployment.

This framework incorporated weighting each variable based on research and description. To address this uncertainty within the framework, a Monte Carlo simulation was added consisting of 10,000 randomised scenarios. This analysis evaluated state ranking, winner frequency, top-three winner frequency, and competitiveness volatility to identify which states remain attractive to corporations under a wide range of assumptions. The results indicate that Texas consistently was shown as the strongest due to its robust location from its low energy pricing, large generation capacity, favorable interconnection conditions, and strong potential for behind-the-meter systems.

The paper also evaluates behind-the-meter power system strategies using simplified models for cost related to grid electricity, natural gas generation, solar, and a hybrid power architecture. The results suggest that optimal power strategies vary drastically across regions and that the hybrid systems may provide the most attractive balance of both cost and reliability across scenarios.

Outside of the regional competitive analysis, the framework projects national electricity demand growth in order to gauge how much infrastructure will be needed by 2030, while exploring the local economic impacts of AI infrastructure deployment for residents. The community-level analysis suggests that while large energy infrastructure projects may increase electricity prices for local residents along with multiple other externalities (sound, temperature, water usage), the projected tax revenue may be sufficient to fund mitigation programs that could offset the economic impacts while leaving reserves to deal with other externalities.

The findings suggest that the future of AI will increasingly be decided by power economics, transmission constraints, and access to energy rather than technological progression of the AI model itself. Results indicate that AI infrastructure deployment is simply a computing problem, but an energy and community planning challenge. The research framework provides a

base for evaluating future decisions around AI infrastructure investment, highlighting areas for further research in power systems, reliability, and economics.

1. Introduction

Artificial intelligence is experiencing rapid growth and is currently the largest driver of new computing related infrastructure investment in the world. Recent advancements in large language models, inference systems, and AI applications in all sectors have increased demand for data centers along with the electricity needed to power them. Companies such as OpenAI, Anthropic, Microsoft, Amazon, Google, Meta, and NVIDIA continue to invest in expanding model capabilities, creating the demand for the infrastructure to train and deploy them.

AI infrastructure growth is not just constrained by computing hardware, but also by physical energy systems. The data centers that power them require significant amounts of electricity, transmission capabilities, cooling systems, and generation sources. As a result of this, access to profit maximising and reliable power is becoming an extremely important factor in determining where this infrastructure is constructed.

Much of current research on AI infrastructure is based around model performance, semiconductor manufacturing, environmental impacts, or just aggregate electricity demand estimations. While these are topics that are important and I have researched in other papers, less attention has been given to the significant factor that geography places a large role in where the infrastructure is deployed. States differ in many conditions such as electricity pricing, generation capacity, access to renewable energy, transmission constraints, and existing infrastructure ecosystems. The key differences may create variation in each region's ability to attract future investment.

Growing electricity demand in turn raised concerns around grid reliability, community impact, and sufficient planning. Policymakers, utilities, and developers are raising questions as to how AI infrastructure should be governed, powered, and how communities are being affected. This gives rise to alternate strategies in location and power generation.

This paper uses a quantitative framework for evaluating AI infrastructure competitiveness across the United States. Opposed to attempting to predict the exact location of future data center development, the goal is to compare each state using a consistent and normalised set of variables that would likely influence investment decisions. The framework includes a state level competitiveness analysis, Monte Carlo simulation, behind-the-meter power economics, national electricity outlook scenarios, and community impact modeling to evaluate where and what kind of energy infrastructure may be effective.

The results of the model suggest that power economics, transmission constraints, and accessible energy may play an increasing role in determining the future of the competitiveness of the AI infrastructure market. Beyond this, the findings indicate that development is not simply constrained by a computing problem. Understanding the relationship may become critical for firms, policymakers, utilities, and investors seeking to support the next generation of AI growth.

2. Methodology

2.1 Research Question

The research evaluates how electricity markets, infrastructure specifics, and power constraints influence AI infrastructure competitiveness across the United States. The objective of the model is not to predict the exact location of future data centers and related infrastructure development but rather to construct a framework that evaluates the relative attractiveness of states for future AI infrastructure investment.

2.2 State Selection

The study focuses on fifteen states that are frequently discussed in the sources as potential future or current markets for AI infrastructure at large scale:
Texas
Virginia
Arizona
California
Nevada
Georgia
Ohio
Oregon
Washington
North Carolina
Utah
Tennessee
Indiana
Pennsylvania
Illinois

These states were selected based on the following qualities:
1. Data center ecosystems
2. Energy resources
3. Infrastructure capacity
4. Overall relevance to ongoing AI infrastructure expansion

2.3 Variable Construction

Version one of the framework uses six variables.

Electricity Price

Industrial electricity prices were obtained from the U.S. Energy Information Administration (EIA). Lower electricity prices are assumed to increase infrastructure competitiveness.

Grid Capacity

Installed electricity generation capacity was collected from EIA state-level data. Higher capacity is assumed to improve the ability to support future infrastructure growth.

Renewable Energy Share

The percentage of electricity generation derived from renewable resources was collected from EIA energy data. Higher renewable penetration may improve long-term sustainability and energy availability due to state laws, lower cost, and supply.

Ecosystem Score

An ecosystem score was constructed using existing data center counts obtained from DataCenterMap. States with larger infrastructure ecosystems are assumed to benefit from existing labor pools, suppliers, supporting infrastructure and economies of scale.

Behind-the-Meter Potential

A behind-the-meter score was created using relative costs of on-site power generation compared with grid electricity. The objective was to estimate the attractiveness of alternative power strategies under different regional conditions based on comparing grid and on-site generation costs.

Interconnection Friction Score

Because consistent state-level interconnection queue duration data was not available, a regional proxy approach was used from sources. States were assigned friction scores based on the relative difficulty of transmission interconnection within their primary interconnection region. Information was derived from reports published by Lawrence Berkeley National Laboratory (LBNL), PJM, CAISO, MISO, and ERCOT.

2.4 Regional Competitiveness Framework

Each variable above was normalized to allow for comparison across states with different scales and units of measurement so that they could be unified within the model.

The normalized variables were combined into a weighted competitiveness score representing the relative attractiveness of each state for future AI infrastructure deployment, best shown in the heatmap visual.

Weighting assumptions were informed by industry reports, infrastructure literature, and observed characteristics of existing AI infrastructure development. This objective weighting is further tested and addressed in the Monte Carlo simulations.

$C_i = 0.25P_i + 0.25K_i + 0.20R_i + 0.20E_i + 0.10I_i$

Where:

P_i = Electricity Price Score

K_i = Grid Capacity Score

R_i = Renewable Energy Score

E_i = Data Center Ecosystem Score

I_i = Interconnection Score=

Each component score is normalized using min-max scaling:

$x'_i = (x_i - \min(x)) / (\max(x) - \min(x))$

2.5 Monte Carlo Robustness Analysis

To address uncertainty in weighting assumptions, the framework incorporates a Monte Carlo simulation consisting of 10,000 randomized scenarios to test possible extremes or objective variations in weighting to determine if the regional competitiveness framework gave preference to certain states.

For each simulation:

1. Variable weights were randomly generated using a Dirichlet distribution.
2. Competitiveness scores were recalculated.
3. States were ranked according to their simulated score.

The simulation was used to estimate:

- Winner Frequency
- Top-Three Frequency
- Median Competitiveness
- Competitiveness Volatility

This approach evaluates whether state rankings remain stable under a broad range of alternative weighting assumptions including extreme scenarios.

$w^{(s)} \sim \text{Dirichlet}(1,1,1,1,1)$

$C_i^{(s)} = \Sigma(w_j^{(s)} \times x_{ij})$

$WF_i = 100 \times (1/N) \times \Sigma\, I(rank_i^{(s)} = 1)$

$\rho_i = \text{Median}(C_i) / \sigma_i$

Where:

WF_i = Winner Frequency

ρ_i = Robustness Score

σ_i = Standard Deviation of Simulated Scores

N = 10,000 Simulations

2.6 Behind-the-Meter Economics Framework

Another framework was developed to research alternative electrical power sourcing for AI infrastructure deployment

6 of the high performing states were used:

- Texas
- Virginia
- Arizona
- California
- Nevada
- Washington

The framework evaluates:

1. Grid Power

Electricity purchased entirely from the grid using state-level industrial electricity prices.

2. Natural Gas Generation

Behind-the-meter generation using state-level natural gas prices and simplified generation cost assumptions.

3. Solar Generation

Utility-scale solar generation using capacity factors estimated from NREL PVWatts simulations.

4. Hybrid Systems

A simplified hybrid architecture combining solar generation, battery storage, and natural gas backup generation. This is the most realistic model since it takes into account a variable the framework does not use, reliability. Reliability will be factored in later research for version two in this specific sector of AI infrastructure.

The objective is to identify which power architecture minimizes electricity costs under different regional conditions.

2.7 National Power Outlook Framework

A national power outlook model was developed to compare projected electricity demand growth with planned generation capacity additions through 2030. This was added to project how much infrastructure will be needed in coming years.

Three scenarios were evaluated:

- Low Growth (10% Growth)
- Baseline Growth (20% Growth)
- High Growth (35% Growth)

The framework compares projected demand growth with estimated generation additions under varying assumptions regarding project completion rates and AI-driven electricity demand. Using three outcomes gave a more realistic range of projections.

$D_{2030} = D_0(1 + g)$

$K_{2030} = K_0 + Pr$

Where:

D_0 = Current National Electricity Demand

K_0 = Current National Generation Capacity

P = Planned Capacity Additions

g = Demand Growth Rate

r = Capacity Realization Rate

2.8 Community Impact Framework

A final framework was developed to evaluate the local impacts of AI infrastructure deployment. This would affect residents of towns where the infrastructure is deployed.

The analysis considers:

- Local Electricity Cost Impacts
- Job Creation
- Tax Revenue Generation
- Construction Activity
- Noise
- Land Use Impacts
- Water Consumption

The framework estimates whether local tax revenues generated by infrastructure projects may be sufficient to offset resident electricity costs and support mitigation programs for other variables designed to reduce community burdens.

$RC_{household}$ = Monthly Bill × 12 × (q/100)

$RC_{community}$ = $RC_{household}$ × Households

M = $RC_{community}$

Surplus = Tax Revenue - $RC_{community}$

Where:

$RC_{household}$ = Annual Household Electricity Cost Increase

$RC_{community}$ = Total Community Electricity Cost Increase

M = Required Mitigation Fund

Surplus = Net Fiscal Benefit After Resident Offsets

3. Results

3.1 Regional Competitiveness Results

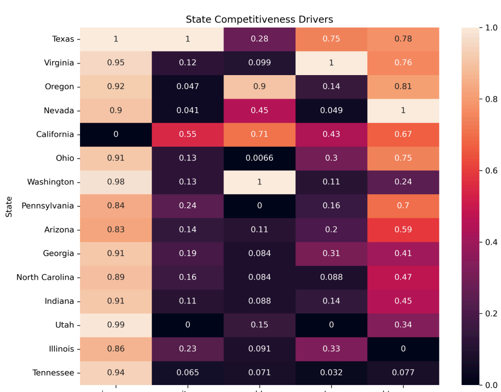
The regional competitiveness framework identified the significant differences found in the attractiveness of states for future AI infrastructure deployment. Using the baseline weight assumptions, Texas was shown to have the highest overall competitiveness score. Texas was followed by Virginia, Arizona, Washington, and Nevada which also performed strongly within the framework.

Texas benefitted from a combination of low industrial electricity prices, a large generation capacity, established infrastructure ecosystem, favorable interconnection conditions, and a relatively strong behind-the-meter potential. This combination of variables allowed Texas to outperform other states across multiple categories opposed to relying on a single advantage.

States with strong renewable energy resources often performed above average in the sustainability-related variables, but were also constrained by higher electricity prices, lower infrastructure density, or transmission limitations. Conversely, this was also seen from states with lower electricity pricing who had worse renewable energy penetration scores. This interestingly shows a clear inverse between the two variables and is something developers and policy makers should take into account.

Overall, the results suggest that AI infrastructure competitiveness is determined by a combination of energy, infrastructure, and ecosystem factors rather than any single variable alone.

Figure 1. Regional Competitiveness Scores



Source: Author calculations.

3.2 Monte Carlo Robustness Results

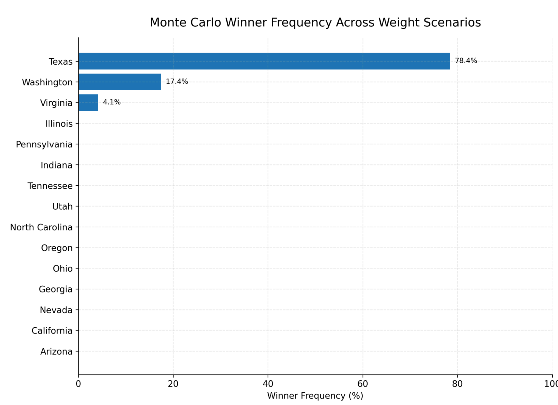
To evaluate the sensitivity of the framework to weighting assumptions, a Monte Carlo simulation consisting of 10,000 randomized scenarios was integrated into the framework.

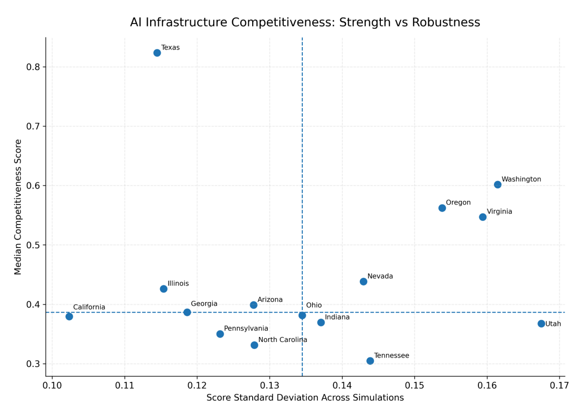
Texas emerged as the most robust state across simulations, achieving the highest winner frequency and consistently appearing among the top-ranked locations. This suggests that Texas performs well under a broad range of potential decision-making priorities rather than only under the baseline weighting assumptions. This supports the baseline weighting, and also further shows the advantage of being strong across multiple variables.

Several states displayed strong average competitiveness but lower robustness. These states performed well when specific variables received greater emphasis but were more sensitive to changes in weighting assumptions. This is still valuable information for certain developers or in extreme cases.

The simulation results indicate that while individual rankings may shift under alternative assumptions, a small group of states (1.Texas, 2. Washington, 3. Virginia) consistently appears among the most attractive locations for future AI infrastructure development.

Figure 2 and 3. Monte Carlo Winner Frequency





Source: Author calculations.

3.3 Behind-the-Meter Economics Results

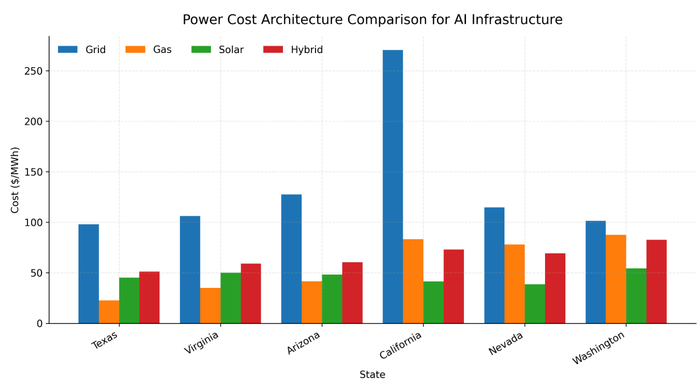
The behind-the-meter economics framework revealed variation in the relative attractiveness of alternative power architectures based on region.

Natural gas generation performed well in regions with relatively low fuel costs and existing energy infrastructure. Solar generation became increasingly attractive in states with strong solar resources and favorable capacity conditions.

Hybrid systems combining solar generation, battery storage, and natural gas backup frequently produced the most balanced outcomes. Although not always the lowest-cost option, hybrid systems offer improved reliability and flexibility compared with single-source architectures, a variable that was not factored in during version one of testing and should be weighed in.

The results suggest that future AI infrastructure deployment strategies may become increasingly location-specific as developers seek to optimize both electricity costs and reliability requirements.

Figure 4. AI Infrastructure Power Cost Comparison



Source: Author calculations.

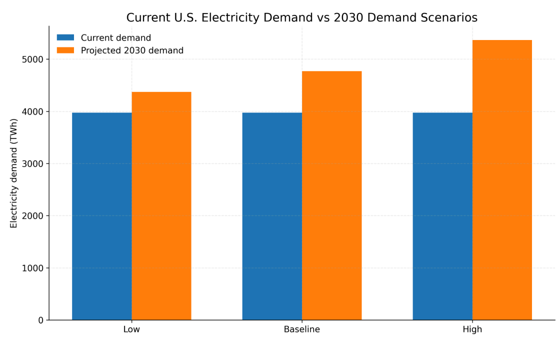
3.4 National Power Outlook Results

The national power outlook framework evaluated whether planned U.S. generation additions appear sufficient to support projected electricity demand growth through 2030. This could be used to determine how much infrastructure would be needed to be used with the other tests that were for determining location.

Across all scenarios, electricity demand increased substantially relative to current levels. While planned generation additions reduce some of the projected supply pressure, the amount of future demand growth remains highly dependent on how much AI is adopted by other industries and is too early in the market to exactly determine.

The analysis suggests that generation expansion alone may not fully resolve future infrastructure constraints. Transmission investment, interconnection reform, and alternative power strategies may become increasingly important components of future infrastructure planning.

Figure 5. Future Electricity Demand Projections



Source: Author calculations.

3.5 Community Impact Results

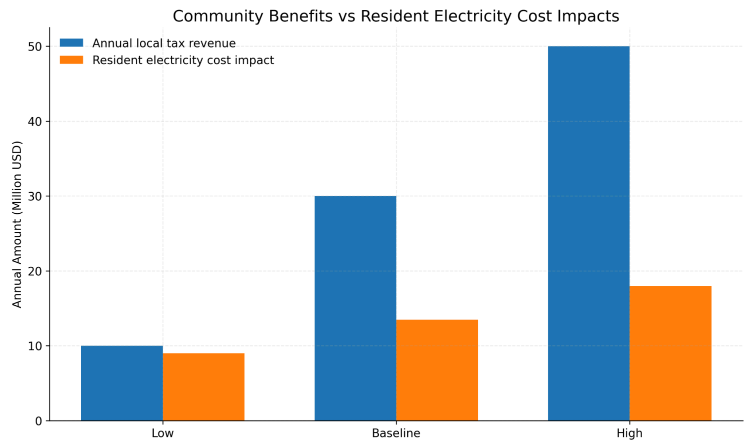
The community impact framework evaluated both the benefits and costs associated with large-scale AI infrastructure deployment for use in determining policy and deployment location based factors for firms.

Potential externalities include increased electricity demand, construction disruption, traffic, noise, land-use impacts, and additional pressure on local infrastructure systems, which could lead to reliability concerns. However, infrastructure projects also generate economic benefits through tax revenue, employment opportunities, and investment activity.

Under baseline assumptions, estimated local tax revenues exceeded the modeled cost of offsetting residential electricity bill increases. This finding suggests that targeted mitigation programs may allow communities to share more directly in the economic benefits generated by AI infrastructure development through subsidizing electricity cost increases for residents and funding for other dealing with other problems (noise, traffic, etc).

While the framework uses simplified assumptions, the results indicate that community opposition may be influenced not only by the magnitude of local impacts but also by how the economic benefits of infrastructure projects are distributed among residents. Policy makers should take this into account when dealing with the negative externalities that come into play with AI growth. Firms should also view this as places to allocate funding for lessening local resident cost and resentment.

Figure 6. Community Tradeoffs



Source: Author calculations.

4. Discussion

After running the models and framework, the results suggest that the future geography of AI infrastructure may be determined more and more by energy economics rather than AI model factors alone. While historical data center development often emphasized network connectivity, tax incentives, and proximity to major city regions, the growing electricity requirements of AI systems may increase power availability and energy costs to primary strategic considerations.

One of the most significant findings of the framework is the consistent performance of Texas across both the baseline competitiveness model and the Monte Carlo robustness analysis (winning the tests almost 80% percent of the time, or top 3 almost 100% of the simulated weightings). Texas benefits from a combination of low electricity prices, substantial generation capacity, favorable interconnection conditions, and an established infrastructure ecosystem. On top of this, Texas remained highly competitive across a broad range of weighting assumptions, from extremes to uniformed weightings, suggesting that its attractiveness is not dependent on any single advantage. This robustness may help explain why many recent infrastructure announcements have been concentrated within the state. A factor not discussed in the framework is region size, something Texas benefits from, allowing distance between data centers and decreasing chances for failures, an attribute a company like Amazon (AWS) deems very important. Using the data from the Monte Carlo simulations shows that even under uncertainty, some regions may be better than others.

The results also highlight the growing importance of transmission and interconnection constraints. Historically, infrastructure planning discussions often focused on the availability of generation resources. However, the ability to connect new loads to the grid may increasingly become a limiting factor, or at very least something to take into account with the need for quick growth. As electricity demand from AI infrastructure grows, transmission bottlenecks and lengthy interconnection queues may influence deployment decisions as much as electricity prices themselves. This finding suggests that transmission policy and grid modernization may play a significant role in shaping future AI infrastructure expansion.

The behind-the-meter analysis further suggests that future infrastructure developers may rely increasingly on alternative power architectures opposed to grid use based on region. While grid electricity remains attractive in many regions, the combination of rising demand, transmission constraints, and reliability requirements may encourage greater use of natural gas generation, solar generation, battery storage, and hybrid systems to combat this. Hybrid architectures were particularly notable because they provide a balance between cost, reliability, and operational flexibility. As AI infrastructure scales, power strategy may become an important competitive advantage.

The national power outlook analysis determines the expected demand for AI related data centers by 2030. Planned generation additions are large, but the results suggest that future outcomes remain highly sensitive to project completion rates and assumptions regarding AI adoption, factors that are uncertain and must be accounted for. Even under optimistic capacity expansion scenarios, transmission infrastructure and power delivery may still remain important constraints. This finding reinforces the importance of long-term infrastructure planning and investment to account for extreme scenarios.

Finally, the community impact framework suggests that local oppositions to infrastructure development are justified, but can be mitigated. The analysis indicates that many of the economic benefits generated by large-scale infrastructure investments may exceed the costs imposed on residents from increased electrical demand without adequate supply. As a result, community acceptance may depend less on eliminating impacts entirely, but more on distributing benefits effectively from developers or policy makers. Programs that offset residential electricity costs or fund local improvements may represent practical mechanisms for aligning infrastructure development with community interests. This also gives more rise to behind-the-meter strategies, involving more job creation and mitigating higher electrical costs for residents.

The results suggest that AI infrastructure development is becoming increasingly dependent on energy systems, transmission networks, and regional economic policy. Firms that successfully combine infrastructure planning, power strategy, and local community factors may be best positioned to support the next generation of AI growth.

5. Limitations

This framework should be interpreted as a comparative decision-support model rather than an exact forecasting model. There are still limitations in the version one model used.

First, the regional competitiveness framework relies on normalized state-level variables. While this allows comparison across states, it may not fully capture local differences within each state, such as specific utility territories, transmission constraints, land availability, water access, or permitting conditions.

Second, the interconnection friction score is based on a regional proxy rather than direct state-level queue duration data. Because consistent state-level interconnection data was not available, the model uses information from interconnection reports to estimate relative friction.

Third, the behind-the-meter economics model is simplified. It compares grid, natural gas, solar, and hybrid power architectures using broad cost assumptions, but it does not factor in reliability, a main advantage of hybrid or gas strategies.

Fourth, the national power outlook uses scenario assumptions rather than a detailed capacity expansion model. This should be used for projecting direction rather than exact amounts.

Finally, the community impact framework monetizes residential electricity cost impacts more directly than other local burdens. Noise, traffic, water use, land use, and visual impacts are included qualitatively or through simplified scoring, but they are not fully converted into monetary welfare estimates, although leftover funds after returning local electrical markets to equilibrium appear to be enough in many scenarios.

Future research could improve the framework by incorporating state-level interconnection queue data, utility-territory analysis, reliability metrics, water constraints, nuclear and geothermal scenarios, and more detailed community cost-benefit estimates.

6. Conclusion

Artificial intelligence is demanding some of the largest infrastructure growth in modern economic history. As AI model capabilities continue to progress, the demand for computing infrastructure, electricity, transmission capacity, and supporting energy resources will only grow in need. Understanding where this infrastructure is likely to be built, how it can be powered, and how local communities near it may be affected represents an increasingly important challenge for firms, utilities, policymakers, and investors.

This paper developed a quantitative framework for evaluating AI infrastructure competitiveness across the United States. By combining state-level energy, infrastructure, and ecosystem variables with Monte Carlo simulation, behind-the-meter power economics, national electricity outlook scenarios, and community impact analysis, the framework creates a structured way for comparing future infrastructure opportunities and constraints.

The model's results suggest that power economics, grid capacity, and interconnection conditions are becoming increasingly important in determining infrastructure competitiveness. Across varying model weights, Texas consistently emerged as the strongest overall deployment location due to its low electricity prices, large generation capacity, favorable infrastructure conditions, and strong robustness across alternative scenarios.

The analysis also covers the growing importance of alternative power strategies. Behind-the-meter on site generation, hybrid energy systems, and transmission investment may become increasingly important as AI electricity demand grows. At the same time, the community impact framework suggests that many of the economic benefits generated by infrastructure projects may be sufficient to support mitigation programs that reduce negative externalities while preserving economic growth.

The findings show that AI infrastructure development is no longer simply a computing challenge. It is increasingly an energy, infrastructure, and community planning challenge. Firms that successfully integrate power strategy, infrastructure economics, and community engagement may be best positioned to support the next generation of AI growth.

While the framework presented here is simplified and has several limitations, it provides a foundation for future research into AI infrastructure deployment, energy system planning, and infrastructure investment decision-making. As AI adoption grows, understanding the relationship between compute, power, and geography may become increasingly important within the industry.

References

Amazon Web Services. "AWS Sustainability." Amazon Web Services, https://sustainability.aboutamazon.com/environment/the-cloud?energyType=true.

California Independent System Operator. "Interconnection Queue Management and Transmission Planning Reports." CAISO, 2025.

Data Center Map. "Data Center Locations in the United States." Data Center Map, https://www.datacentermap.com/usa/.

Electric Power Research Institute. "Powering Intelligence: Analyzing Artificial Intelligence and Data Center Energy Consumption." EPRI, https://powering-intelligence.epri.com/executive-summary.html.

Electric Reliability Council of Texas. "Generation Interconnection Status Reports." ERCOT, 2025.

Google. "Data Center Sustainability." Google Data Centers, https://www.google.com/about/datacenters/sustainability/.

International Energy Agency. *Energy and AI*. International Energy Agency, https://www.iea.org/reports/energy-and-ai.

Lawrence Berkeley National Laboratory. *2024 LBNL Data Center Energy Usage Report*. Lawrence Berkeley National Laboratory, https://eta.lbl.gov/publications/2024-lbnl-data-center-energy-usage-report.

Lawrence Berkeley National Laboratory. *Interconnection Queue Data and Reports*. Lawrence Berkeley National Laboratory, 2025.

Meta. "Data Center Sustainability." Meta Sustainability, https://sustainability.atmeta.com/data-centers/.

Microsoft. "Microsoft Sustainability." Microsoft, https://www.microsoft.com/en-us/sustainability.

Midcontinent Independent System Operator. "Generator Interconnection Queue Reports." MISO, 2025.

National Renewable Energy Laboratory. "PVWatts Calculator." NREL, https://pvwatts.nrel.gov/.

NVIDIA. "NVIDIA H100 Tensor Core GPU." NVIDIA, https://www.nvidia.com/en-us/data-center/h100/.

NVIDIA. "NVIDIA Blackwell Architecture." NVIDIA, https://www.nvidia.com/en-us/data-center/technologies/blackwell-architecture/.

PJM Interconnection. "Interconnection Process Reform and Queue Statistics." PJM Interconnection, 2025.

Rand, Joseph, et al. *Queued Up: 2025 Edition—Characteristics of Power Plants Seeking Transmission Interconnection as of the End of 2024*. Lawrence Berkeley National Laboratory, Dec. 2025, https://emp.lbl.gov/queues.

United States Department of Energy. "Clean Energy Resources to Meet Data Center Electricity Demand." U.S. Department of Energy, https://www.energy.gov/oe/clean-energy-resources-meet-data-center-electricity-demand

United States Energy Information Administration. *Electric Power Monthly*. U.S. Energy Information Administration, https://www.eia.gov/electricity/monthly/.

United States Energy Information Administration. "Electricity Data Browser." U.S. Energy Information Administration, https://www.eia.gov/electricity/data.php.

United States Energy Information Administration. "U.S. States." U.S. Energy Information Administration, https://www.eia.gov/state/.

Uptime Institute. "Power Usage Effectiveness (PUE)." Uptime Institute, https://uptimeinstitute.com/pue.